

NewsInsight2.0: An Enhanced Version Integrating Large Language Model-based Query Optimization with Advanced Temporal Mechanisms

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Abstract. We introduce NewsInsight2.0, a cutting-edge evolution of the NewsInsight system, specifically designed for the Ho Chi Minh AI Challenge 2024. Building on the strengths of ViewsInsight, NewsInsight2.0 addresses previous performance limitations with significant enhancements. Our optimized architecture is tailored to deliver exceptional search capabilities for AIC 2024. At its core, NewsInsight2.0 leverages the CLIP (Contrastive Language-Image Pre-training) model, trained on a vast dataset of 5 billion parameters (DFN-5B). Additionally, we have refined our temporal query mechanism with a more efficient algorithm and an intuitive user interface. Furthermore, NewsInsight2.0 features an automatic query generator powered by open-source large language models, streamlining the process of optimizing user input queries.

Keywords: Lifelog · Video Retrieval System · Automatic Retrieval System · Interactive Retrieval System · Automatic Retrieval System · AI-based Assistance · Large Language Models

1 Introduction

The AI Challenge 2024 is a benchmarking event focused on visual data retrieval, aiming to advance real-time event extraction from videos in large databases. This challenge mirrors international competitions like the Lifelog Search Challenge (LSC) [5] and Video Browser Showdown (VBS) [10] and emphasizes practical AI-driven solutions. Participants develop systems to identify specific events from videos based on descriptions, gaining industry insights and practical skills. The challenge includes three query types:

1. **Textual Known-Item Search (Textual-KIS):** Participants locate video segments from detailed, natural language descriptions. The full description is provided in the preliminary round, while in the final, it is revealed incrementally within a 5-minute window.

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2. **Video Known-Item Search (Video-KIS):** In the final round, participants view a brief clip (up to 20 seconds) and use alternative methods, like text or sketches, to locate the matching event, because recording the video is prohibited.
3. **Question and Answering (Q & A):** Similar to Textual-KIS, participants answer questions about a described event in the video. Full descriptions are given in the preliminary round, while in the final, they are progressively revealed.

In this research, we present NewsInsight2.0, an enhancement of NewsInsight [20], an enhanced retrieval system powered by large language models designed for the AI Challenge 2024. In summary, this paper makes the following key contributions:

- The utilization of Contrastive Language-Image Pre-training (CLIP) [16], trained on DFN-5B. Data Filtering Networks (DFNs) [2] are compact systems developed to filter large amounts of uncurated data. The model was trained on 5 billion images, chosen from a pool of about 43 billion unfiltered image-text pairs, with 12.8 billion taken from CommonPool-12.8B [3] and the other 30 billion additional public pairs.
- The temporal query mechanism is improved by implementing a more efficient algorithm and introducing a more user-friendly interface for this feature.
- For dealing with the Vietnamese description of the event description, we propose an integration of VinAI Translate [12], a state-of-the-art text translation model for Vietnamese to English.
- With the help of modern large language models, a query optimization system is proposed to provide suggestions for improving the quality of the prompt, helping to enhance the retrieval result for video events.

The structure of this paper is organized as follows: Section 2 provides a brief overview of existing retrieval systems developed for video event search. Section 3 introduces an overview and user interface of NewsInsight2.0. Enhancement features and components of our system are presented in section 4. Some experiments and results of our system are demonstrated in section ???. This paper ends with section 5, where there is a summary of our findings and a discussion of future work.

2 Related Work

The field of video retrieval has experienced significant enhancements in recent years, resulting in the organization of competitions focused on developing interactive video retrieval systems, for example, the Lifelog Search Challenge (LSC) [5] or the Video Browser Showdown (VBS) [10]. Different systems have been introduced, with the aim of effectively locating specific images, from an enormous collection of videos, within a limited timeframe and based on a user’s query.

Various approaches, including Memoria [4], LifeInsight [13], and LifeSeeker [8], have been introduced with multiple search modalities based on concepts.

Other systems, such as lifeXplore [15], used convolutional neural networks (CNNs) and traditional object detection models for content analysis. These systems primarily used Database Management Systems (DMBS) or Elasticsearch for aligning user queries with visual concepts and metadata. These systems have shown noteworthy improvements in zero-shot image retrieval, compared to their previous versions.

Concept-based search is a method used to enhance image retrieval by tagging images with visual markers or concepts, such as detected objects, image captions, or texts extracted via Optical Character Recognition (OCR). These annotations, along with other metadata, are utilized for the matching between the images and the input keywords from users. Moreover, concept-based search also incorporates Natural Language Processing (NLP) techniques to further refine the retrieval results by extracting key concepts from textual information associated with the image, as observed in systems like V-FIRST [7].

Semantic search enhances image and video retrieval by emphasizing the interpretation of meaning and context within visual content, leading to more accurate results than traditional methods, which focus on surface features only. Contemporary systems leverage vision-language pre-trained models, such as CLIP [16] to enhance the performance in tasks such as zero-shot image-text retrieval. Examples of this can be seen in the work of Lokoč et al [11], or diveXplore [17]. Additionally, advanced models like BLIP [9] have further improved the retrieval performance by providing a deeper understanding of semantic content.

Moreover, Heller et al. [6], with the VITRIVR system shows the effectiveness of applying the temporal query mechanism in handling complex queries, which include temporal data. In the 2024 Video Browser Showdown, ViewsInsight [19] employed this power and achieved an outstanding performance.

At the 2023 AI Challenge, we published NewsInsight [20], an interactive video retrieval system, designed to efficiently locate video frames based on user inputs. Although the system showed several advantages, it required users to manually optimize their prompts. This has led to the creation of an automatic retrieval system, in which the query optimization will be processed in the background, reducing human efforts while using the system.

3 Overview of NewsInsight2.0

3.1 Overview

In 2023, we introduced NewsInsight [20], a system with notable capabilities in video retrieval. Despite its strengths, the system faced challenges, such as the inclusion of irrelevant frames, text extraction from frames, and the need for manual translation and query optimization. To address these issues, NewsInsight2.0 is built upon NewsInsight, with a query optimization process based on large language models (to be more specific, Meta-Llama3.1 [1], Stanza [14] and T5-base model [18]), and additional advanced temporal mechanisms.

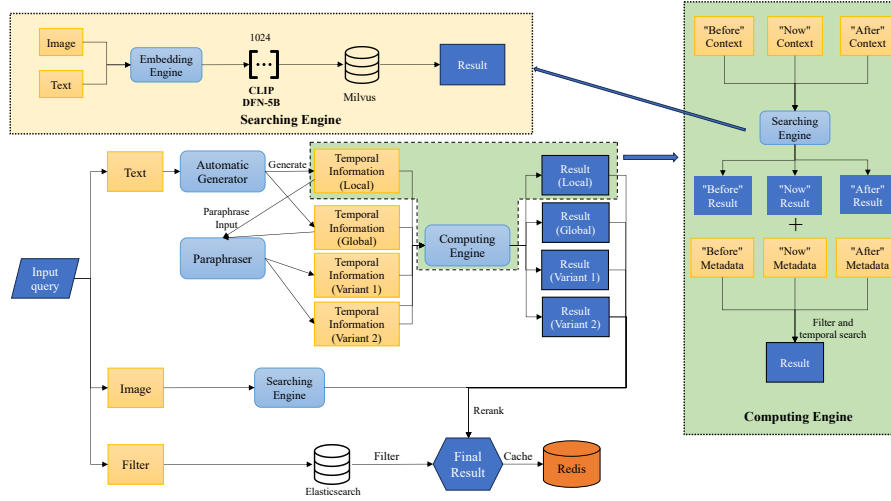


Fig. 1: Overview of NewsInsight2.0.

An overview of NewsInsight2.0 is illustrated in Figure ?? . The system is capable of processing input queries in multiple forms, such as visual images, semantic texts, or filters (by OCR).

For the illustration of the pipeline, we will use the following query, taken from the 2024 Ho Chi Minh City AI Challenge: “News about Christmas in a European city. Decorative lights are hanging above the streets between buildings. There are lights shaped like fish and cube-shaped decorations with a red jewel in the middle. Finally, there is an image of a Christmas tree wrapped with many lights.” The query text is initially processed by an automatic generator that extracts both global and local temporal information. In this instance, the context of the current event is determined without considering preceding or subsequent contexts. The results produced by this automatic query generator are “Intricately designed Christmas lights and decorations, including fish-shaped lights and cube-shaped ornaments with red jewels, adorn the street” for the local context and “A charming European city during the holiday season, with festive lights and decorations illuminating the streets and buildings” for the global context. Besides that, metadata such as the date (Christmas), location (a European city), and concepts (city, fish, red, image, tree, wrapped) are also retrieved. The result of this query is shown in Figure 2.

3.2 User Interface

An improvement to NewsInsight2.0’s UI over the previous version is the new horizontal mode, which displays images in a video-like, horizontal layout. For instance, when a user searches for “xin chào” the results are presented as shown in Figure 3.

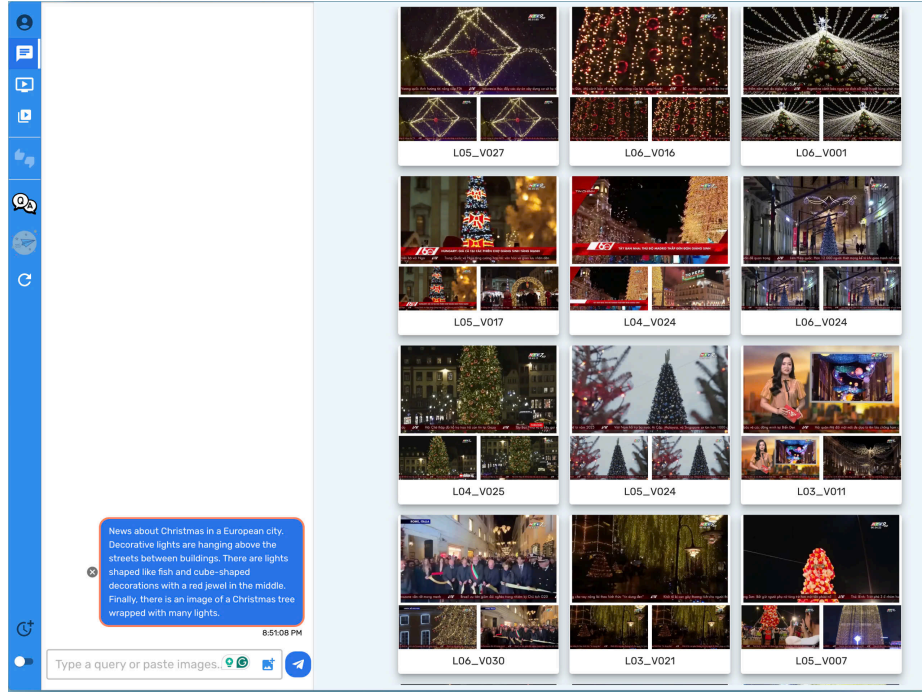


Fig. 2: Illustration of NewsInsight2.0 applied to a query in the 2024 HCMC AI Challenge.

4 Enhancement Features

4.1 Data Preprocessing

In the Ho Chi Minh AI Challenge, removing redundant extracted frames is essential for boosting search performance. Our approach begins with manually selecting representative junk frames (as Fig 4), then applies visual similarity filtering with a specific threshold to detect similar frames. Specifically, for each chosen junk frame, we filter out all related frames that have a visual similarity score above 0.95. This preprocessing step significantly reduces the number of entries inserted into the Milvus vector database, which in turn decreases retrieval time and enhances overall efficiency.

4.2 Automatic Query Generator

The Automatic Query Generator in NewsInsight2.0 incorporates three essential models: Llama 3.1 [1], Stanza [14], and a T5-based model [18]. Each model is vital for enhancing the system’s capability to process and comprehend queries, as shown in Fig 5.

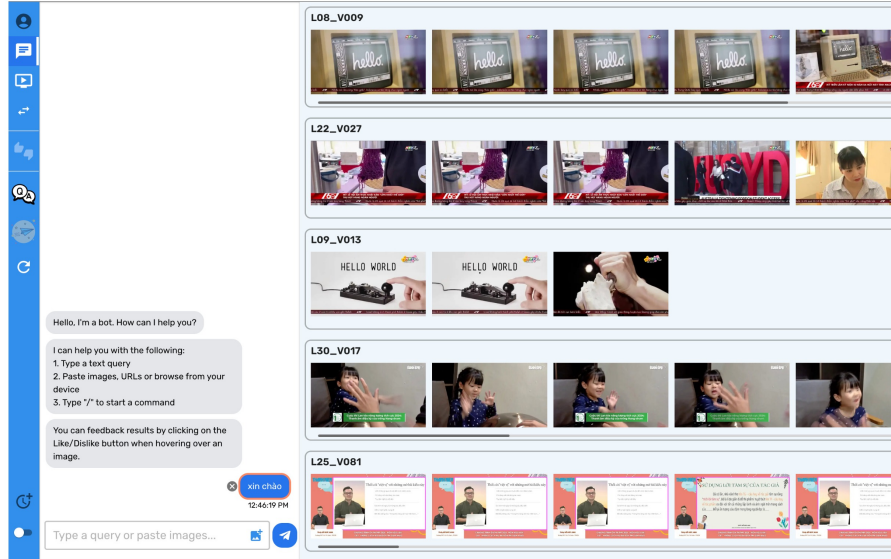


Fig. 3: Horizontal mode of NewsInsight2.0 is split into two sides. The left-hand side includes a query box and toolbar, and the remaining shows results relevant to the input query.



Fig. 4: An example junk frame from provided videos.

- **Llama 3.1** is responsible for extracting temporal information from input queries and breaking it into smaller temporal units. For each unit, it gen-

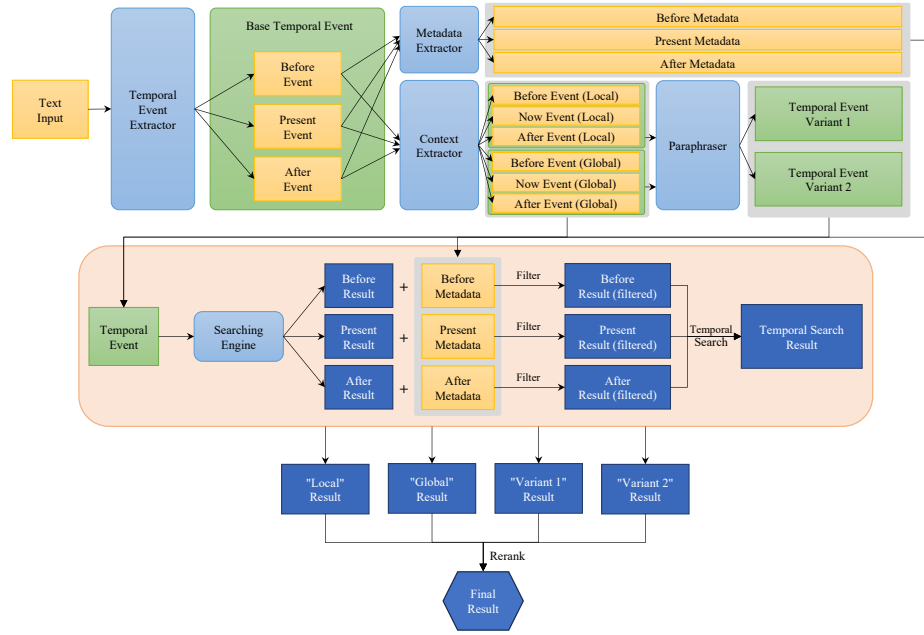


Fig. 5: An overview of the Automatic Query Generator for temporal search.

erates local and context-specific prompts that help contextualize the query. This process not only allows for a deeper understanding of the temporal aspects of the query but also enriches the input for subsequent processing stages.

- **Stanza** functions as the system’s metadata extractor. It evaluates the query to identify key components, such as concepts, locations, and time references, providing both broader context and detailed information. Stanza’s advanced parsing capabilities enable it to manage a variety of input queries, ensuring precise interpretation across a wide array of user inputs.
- Lastly, the **T5-based model** focuses on paraphrasing the context prompts generated by Llama 3.1. Producing alternative phrasing variations helps in minimizing biases during data retrieval and improves the overall accuracy of the query interpretation process.

RAW INPUT: A scene shows a dog wagging its tail. The dog’s head is partially hidden behind a wood-grain door, which we then realize is part of a refrigerator. The next shot shows someone feeding the dog. Following automatic query parsing within NewsInsight2.0 the output will be:

Before:

Local prompting:	A curious dog is peeking out from behind a partially open refrigerator door.
Global prompting:	A typical household kitchen scene with everyday appliances and possibly food items scattered around.
Variants:	The refrigerator door is being guarded by a curious canine who is peeking out from behind it. Common household items, including food and common household tools, gathered in the kitchen.

Now:

Local prompting:	A happy dog expressing excitement or friendliness.
Global prompting:	A scene at home, in a park, or at a pet-friendly location where the dog feels comfortable and at ease.
Variants:	A joyful or playful dog expressing excitement or affection. A place where the dog feels comfortable and safe at home, in a pet-friendly park, or in an environment that is conducive to their pets.

After:

Local prompting:	A person pouring food into the dog's bowl, possibly with the dog sitting nearby, waiting eagerly to eat.
Global prompting:	A quiet domestic scene at home, with the sounds of everyday life in the background.
Variants:	Someone putting food into the dog's bowl, possibly with the animal sitting nearby, eagerly waiting to eat. Home is a serene setting where everyday sounds of life fill the air.

4.3 Temporal Query Mechanism

As mentioned in section 2, “Multi-Stage Queries and Temporal Scoring in VITRIVR” [6] of Heller et al. with the VITRIVR system demonstrated the efficiency of the temporal query mechanism. After that, ViewInsight [19] leveraged this mechanism and achieved high-ranking results in navigating and retrieving related video segments during the 2024 Video Browser Showdown competition.

In this paper, NewsInsight2.0 introduces event-level filtering, which helps to deliver more precise results while reducing redundancy. To utilize this temporal query function, the system requires three inputs from users: the main query and two temporal descriptions that outline events occurring before and after the main event in the query. The algorithm then processes these inputs, extracting relevant information. It then adjusts the score of the “current” query by considering the highest score from the “before” and “after” queries associated with the same video ID. Finally, the algorithm produces a ranked list of search results based on these updated scores. This robust feature is smoothly integrated into the system, allowing it to retrieve relevant results efficiently. This integration enhances the system’s overall functionality, effectiveness, and user-friendliness. The temporal query mechanism is powered by a sophisticated temporal search algorithm, detailed in Algorithm 1.

Algorithm 1 Temporal Search Algorithm

Require: a, b, c, A, C, B

- 1: // Initialize variables
- 2: $A_1 \leftarrow \{\}$
- 3: $C_1 \leftarrow \{\}$
- 4: **for all** b_i in B **do**
- 5: $A_1 \leftarrow$ Subset of A with video id from b_i
- 6: $C_1 \leftarrow$ Subset of C with video id from b_i
- 7: **if** A_1 is empty **then**
- 8: $a_i \leftarrow 0$
- 9: **else**
- 10: $a_i \leftarrow$ Element in A_1 with the highest score
- 11: **end if**
- 12: **if** C_1 is empty **then**
- 13: $c_i \leftarrow 0$
- 14: **else**
- 15: $c_i \leftarrow$ Element in C_1 with the highest score
- 16: **end if**
- 17: Update score of b_i : $b_i.score \leftarrow b_i.score + a_i.score + c_i.score$
- 18: **end for**
- 19: Sort B in descending order based on updated scores
- 20: **return** Sorted B as relevant image results

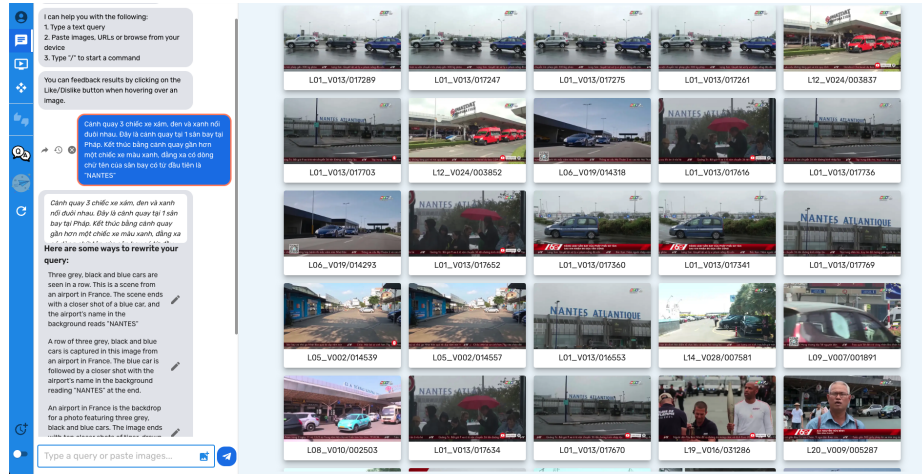


Fig. 6: Use text search to find the expected image.

4.4 Testing

Query: *Footage shows three cars—gray, black, and blue—following each other in line. This scene was filmed at an airport in France. It ends with a closer shot of the blue car, with the name of the airport visible in the distance, starting with the word "NANTES".*

We employ a straightforward text search method, utilizing the image description as our search query. By searching this way, we identify the top-1 candidate image, shown in Figure 6, which is the image that matches the result.

5 Conclusion

In conclusion, the NewsInsight2.0 video event retrieval system is a comprehensive system that uses a variety of mechanisms and features to provide insightful and relevant search results for video event data. It incorporates semantic search mechanisms from state-of-the-art systems and focuses on using spatial information to retrieve relevant information from video event data. The use of the BLIP model for zero-shot image-text retrieval and Elasticsearch for filtering irrelevant images has been shown to improve the precision and recall scores of the system in previous experiments. The integration of visual similarity search functionality and explicit relevance feedback further enhances the accuracy of the search results, while AI-based query description rewriting and visual example generation features provide additional support for end-users. Our system, NewsInsight2.0, has demonstrated its efficacy in retrieving pertinent images corresponding to a given query.

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